

M02 Protected Review and Repair

OPEN ONLY AFTER A SAVED GENUINE ATTEMPT. A corrected query after reading this file is learning evidence, not mastery. Close this file and use a changed retry for mastery.

Review - DE02-01 - Database, table, row, column and query mental model

One valid reference pattern:

```
SELECT DATABASE();  
SELECT COUNT(*) FROM sales_orders;
```

Observed/control target: stage02_retail_lab; 30 rows

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-02 - SELECT, aliases and LIMIT

One valid reference pattern:

```
SELECT order_id, qty, unit_price, qty*unit_price AS gross_sales FROM sales_orders LIMIT 10;
```

Observed/control target: O1001=1700; O1002=1350

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-03 - WHERE and comparison operators

One valid reference pattern:

```
SELECT COUNT(*) FROM sales_orders WHERE status='Completed' AND qty>=2;
```

Observed/control target: 22

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-04 - AND, OR, NOT and parentheses

One valid reference pattern:

```
SELECT COUNT(*) FROM sales_orders WHERE status='Completed' AND (region='East' OR region='North');
```

Observed/control target: 13

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-05 - ORDER BY, DISTINCT and result inspection

One valid reference pattern:

```
SELECT DISTINCT region FROM sales_orders ORDER BY region;
```

Observed/control target: East, North, South, West

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-06 - NULL, IS NULL and COALESCE

One valid reference pattern:

```
SELECT COUNT(*), COUNT(sales_rep) FROM sales_orders;  
SELECT order_id FROM sales_orders WHERE sales_rep IS NULL;
```

Observed/control target: 30 vs 29; O1030

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-07 - Arithmetic expressions and safe calculations

One valid reference pattern:

```
SELECT order_id, qty*unit_price AS gross_sales, qty*unit_price*(1-discount_pct) AS net_sales FROM sales_orders;
```

Observed/control target: O1002 1350 / 1282.50

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-08 - Aggregate functions

One valid reference pattern:

```
SELECT COUNT(*), SUM(qty), AVG(unit_price), SUM(qty*unit_price) FROM sales_orders;
```

Observed/control target: 30 / 105 / ~666.33 / 50000

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-09 - GROUP BY and grouped grain

One valid reference pattern:

```
SELECT region, COUNT(*) orders, SUM(qty*unit_price) gross_sales FROM sales_orders WHERE status='Completed' GROUP BY region;
```

Observed/control target: East 8/12800; North 5/9310; South 7/12010; West 7/10930

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-10 - HAVING after aggregation

One valid reference pattern:

```
SELECT region,SUM(qty) total_qty FROM sales_orders GROUP BY region HAVING SUM(qty)>=25;
```

Observed/control target: East 29; North 27; West 25

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-11 - CASE for business categories

One valid reference pattern:

```
SELECT CASE WHEN qty*unit_price>=2500 THEN 'High' WHEN qty*unit_price>=1500 THEN 'Medium' ELSE 'Small' END bucket, COUNT(*) FROM sales_orders GROUP BY bucket;
```

Observed/control target: High 3; Medium 14; Small 13

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-12 - String and date basics

One valid reference pattern:

```
SELECT order_id,UPPER(TRIM(region)),YEAR(order_date),MONTH(order_date) FROM sales_orders WHERE order_date BETWEEN '2026-07-10' AND '2026-07-12';
```

Observed/control target: 6 rows; year 2026; month 7

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-13 - Simple subqueries and IN/EXISTS awareness

One valid reference pattern:

```
SELECT order_id, qty*unit_price*(1-discount_pct) net_sales FROM sales_orders WHERE status='Completed' AND qty*unit_price*(1-discount_pct) > (SELECT AVG(qty*unit_price*(1-discount_pct)) FROM sales_orders WHERE status='Completed');
```

Observed/control target: 13 rows; average ~1592.02

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Review - DE02-14 - Debugging and validation workflow

One valid reference pattern:

```
SELECT COUNT(*) FROM sales_orders WHERE status='Completed';  
SELECT SUM(qty*unit_price) FROM sales_orders WHERE status='Completed';
```

Observed/control target: 27 Completed rows; Completed gross control = 45050

Repair only the failed concept. Then close this review and complete a changed fresh task. Preserve the original attempt.

Module practical review rubric

Competency	Points	Critical failure
Eligibility/filter logic	15	Wrong population accepted as correct
Grain + grouping	20	Cannot state source/result grain
NULL + calculations	15	NULL treated as zero/known or formula materially wrong
Aggregates/reconciliation	20	Important totals unreconciled
Subquery reasoning	10	Inner query not independently checked
Debug/recovery evidence	10	Random edits / failed attempt erased
Explanation/evidence	10	Cannot explain result or show saved evidence